

FinTech Research Report: Data Analytics in Zerodha

1. Introduction

FinTech, or Financial Technology, refers to the use of technology to provide financial services. Modern FinTech companies generate and process large amounts of data from transactions, market activity, customer accounts, and digital platforms. Data analytics helps these companies convert this data into useful information for customers and business operations.

For this report, Zerodha has been selected as an example of an Indian FinTech platform. Zerodha started operations in 2010 and developed technology-focused platforms for investing and trading. According to Zerodha, its ecosystem now serves more than 1.6 crore clients and contributes more than 15% of Indian retail trading volumes.

2. About Zerodha

Zerodha is an Indian online brokerage and investment platform. It provides access to products such as stocks, derivatives, mutual funds, ETFs, bonds and IPOs. Its major technology platforms include Kite, Console, Coin and Kite Connect API.

Major Platforms

Kite:

Kite is Zerodha's trading platform. It provides streaming market data, charts, market depth, alerts, portfolio information and other tools for analysing financial markets.

Console:

Console is the central dashboard for a Zerodha account. It provides reports, portfolio analytics, trade information, visualisations and tax-related reports.

Coin:

Coin is Zerodha's platform for investing in direct mutual funds.

Kite Connect API:

Kite Connect provides HTTP/JSON APIs that allow developers and businesses to build applications using Zerodha's platform.

3. Role of Data Analytics

Data analytics is important in a financial platform because large amounts of information must be processed accurately and presented clearly.

Zerodha's Console provides an example of analytics being directly exposed to customers. It provides multi-dimensional insights into trades and portfolios through visualisations. It also provides detailed trade breakdowns and performance information.

The platform processes historical trade information to calculate portfolio profit and loss. Zerodha states that Console processes tens of billions of rows of historical trade breakdowns to account for factors such as corporate actions, splits and transfers when calculating P&L.

4. How Analytics Improves Customer Experience

a. Portfolio Analysis

Customers can analyse their investments and view portfolio performance using reports and visualisations. This reduces the need for customers to manually calculate their investment performance.

b. Profit and Loss Calculation

Analytics is used to calculate and present P&L information. Historical trades and corporate actions need to be considered to produce accurate results.

c. Trade Analysis

Console provides detailed information about individual trades and associated charges such as brokerage and STT. This allows customers to understand their trading activity more clearly.

d. Visualisation

Instead of presenting only raw transaction data, Zerodha provides visualisations and analytical views. Console includes tools such as trade-book and P&L heatmaps.

e. Market Information

Kite provides streaming market data, historical data, advanced charts, market-depth information and alerts. These features help users understand market information in a structured way.

5. Analytics for Business Decisions

Data analytics can also support the operation and improvement of a FinTech platform. Data related to transactions, system performance, reports and customer activity can help teams understand how their systems are performing and where improvements are required.

However, it is important to accurately represent Zerodha's approach: Zerodha says it does not use third-party trackers for behavioural profiling and states that it does not use analytics or A/B testing in its apps to drive revenue or engagement. Therefore, the role of analytics at Zerodha should not be described as traditional customer profiling or aggressive marketing.

Instead, a major visible use of analytics is in financial calculations, reporting, portfolio analysis, market-data presentation, and improving the reliability and functionality of its technology platforms.

6. Data Flow Example

A simplified data flow in a FinTech platform can be represented as:

User Activity → Application/API → Data Processing → Database → Analytics → Dashboard

For example:

1. A customer places a transaction through the application.
2. The transaction is processed through the platform's backend systems.
3. Relevant transaction and market information is stored.
4. Data processing systems organise the information.
5. Analytical calculations are performed.
6. The resulting information is displayed through reports, charts or dashboards.

This type of workflow is closely related to the work of Data Analysts because analysts use processed data to identify trends, calculate metrics and create useful reports.

7. Benefits of Data Analytics

Customer Benefits

- Better understanding of portfolio performance
- Easier access to trading reports
- Clear P&L information
- Visual representation of financial information
- Easier tracking of transactions and investments

Business Benefits

- Better monitoring of platform operations
- Accurate financial reporting
- Improved data processing
- Better understanding of system and product performance
- Support for technology and product improvements

8. Importance for Data Analysts

Zerodha demonstrates why Data Analysts are important in FinTech companies. A Data Analyst may work with transaction data, market data, customer-related operational data, financial metrics and platform-performance information.

Typical analytical activities can include:

- Data cleaning
- Data validation
- SQL querying
- Statistical analysis
- Trend analysis
- KPI calculation
- Dashboard development
- Data visualisation

- Report generation

These activities help transform raw financial data into information that can support customers, technology teams and business decisions.

9. Conclusion

Zerodha is an example of how technology and financial services can be combined to create a digital investment platform. Its products use large amounts of financial and transaction data to provide market information, portfolio analysis, reports, P&L calculations and visualisations.

The company's approach also highlights an important point about FinTech analytics: analytics does not always mean customer profiling or targeted marketing. It can also be used for accurate financial calculations, reporting, data visualisation, platform reliability and improving the overall user experience.

Therefore, understanding data analytics, APIs, databases, financial data and dashboards is valuable for Data Analysts working in the FinTech industry.

10. References

1. Zerodha — About the Company
2. Zerodha — Products and Technology Platforms
3. Zerodha — Kite
4. Zerodha — Console

5. Zerodha Support — Console

6. Zerodha — Product and Business Philosophies